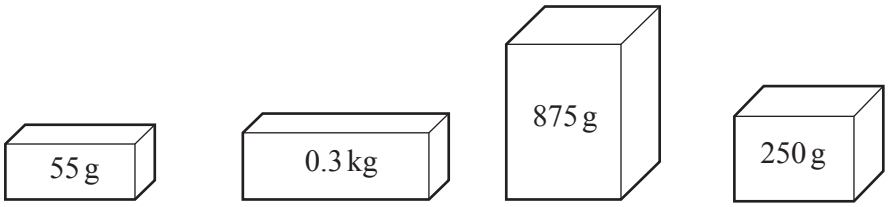
0580_m20_qp_32_Q1

1 Navja works in a post office.

(a) The table shows the costs of sending parcels by post.
The cost depends on the mass, m grams, of the parcel.

Type of parcel	Mass (g)	Cost ($)
Small	$0 < m \leqslant 60$	0.76
Medium	$60 < m \leqslant 100$	0.95
Large	$100 < m \leqslant 250$	2.20
Extra large	$250 < m \leqslant 1000$	5.60

(i) Sai sends each of these four parcels by post.



He pays with a $20 note.

Work out how much change he receives.

$ ..... [4]

(ii) On 1 April, the cost of sending any parcel increases by 5%.

(a) Show that the increase in the cost of sending an **Extra large** parcel is $0.28 .

[1]

(b) Avani says

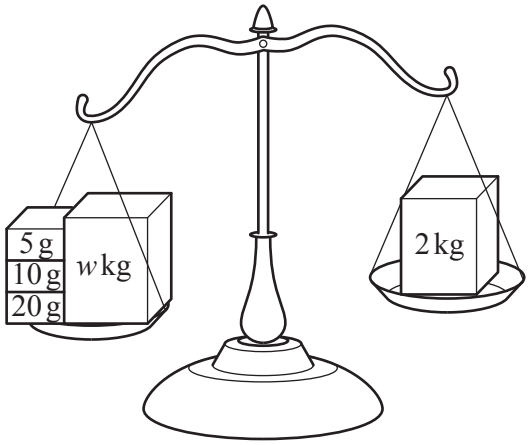
"As the cost of an **Extra large** parcel increases by $0.28 then the cost of a **Large** parcel will also increase by $0.28 to $2.48."

Explain why Avani is incorrect.

.....

..... [1]

(b) **(i)** Navja weighs a parcel with mass w kg on her scales.
She uses the masses shown to balance the scales.



Work out the value of w.

$w =$ ..... [3]

(ii) Sometimes Navja uses an electronic weighing machine.
The machine gives the mass, p kg, of a parcel as 12.4 kg, correct to the nearest 100 g.

Complete this statement about the value of p.

..... $\leqslant p <$ ..... [2]